Find the Hilbert series of the following graded algebras:

(a) $A = k[x_1, \ldots, x_m]$ (where the grading is by degree of polynomials).

(b) $A = k\langle x_1, \ldots, x_m \rangle$ (the grading is by length of words).

(c) A is the exterior (= Grassmann) algebra $\wedge_k[x_1, \ldots, x_m]$ generated over some field k by $x_1, \ldots, x_m$ with the defining relations $x_i x_j + x_j x_i = 0$ and $x_i^2 = 0$ for all i, j (the grading is by degree).

(d) A is the path algebra P_Q of a quiver Q (the grading is defined by $\deg(p_i) = 0$, $\deg(a_h) = 1$).

Hint: The closed answer is written in terms of the adjacency matrix M_Q of Q.

2.9. Lie algebras

Let $\mathfrak{g}$ be a vector space over a field k, and let $[\ ,\ ] : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}$ be a skew-symmetric bilinear map. (That is, $[a, a] = 0$, and hence $[a, b] = -[b, a]$.)

Definition 2.9.1. $(\mathfrak{g}, [\ ,\ ])$ is a **Lie algebra** if $[\ ,\ ]$ satisfies the Jacobi identity

$$\big[[a, b], c\big] + \big[[b, c], a\big] + \big[[c, a], b\big] = 0. \tag{2.9.1}$$

Example 2.9.2. Some examples of Lie algebras are:

(1) Any space $\mathfrak{g}$ with $[\ ,\ ] = 0$ (abelian Lie algebra).

(2) Any associative algebra A with $[a, b] = ab - ba$, in particular, the endomorphism algebra $A = \mathrm{End}(V)$, where V is a vector space. When such an A is regarded as a Lie algebra, it is often denoted by $\mathfrak{gl}(V)$ (general linear Lie algebra).

(3) Any subspace U of an associative algebra A such that $[a, b] \in U$ for all $a, b \in U$.

(4) The space $\mathrm{Der}(A)$ of derivations of an algebra A, i.e. linear maps $D : A \to A$ which satisfy the Leibniz rule:

$$D(ab) = D(a)b + aD(b).$$